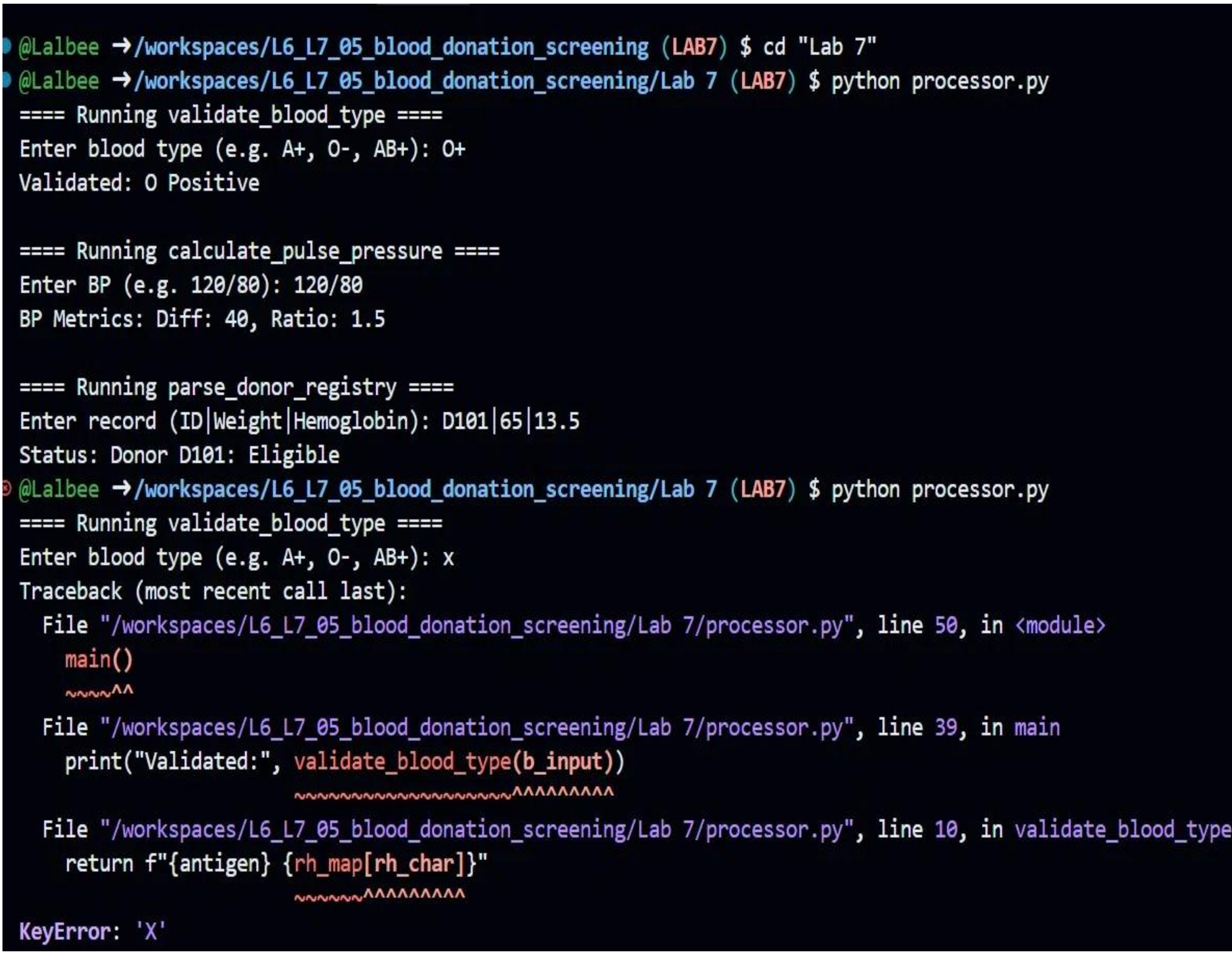
Lab 7: Fuzzing for Input Validation Bugs

NAME :LALBEE

SRN:PES2UG24CS251

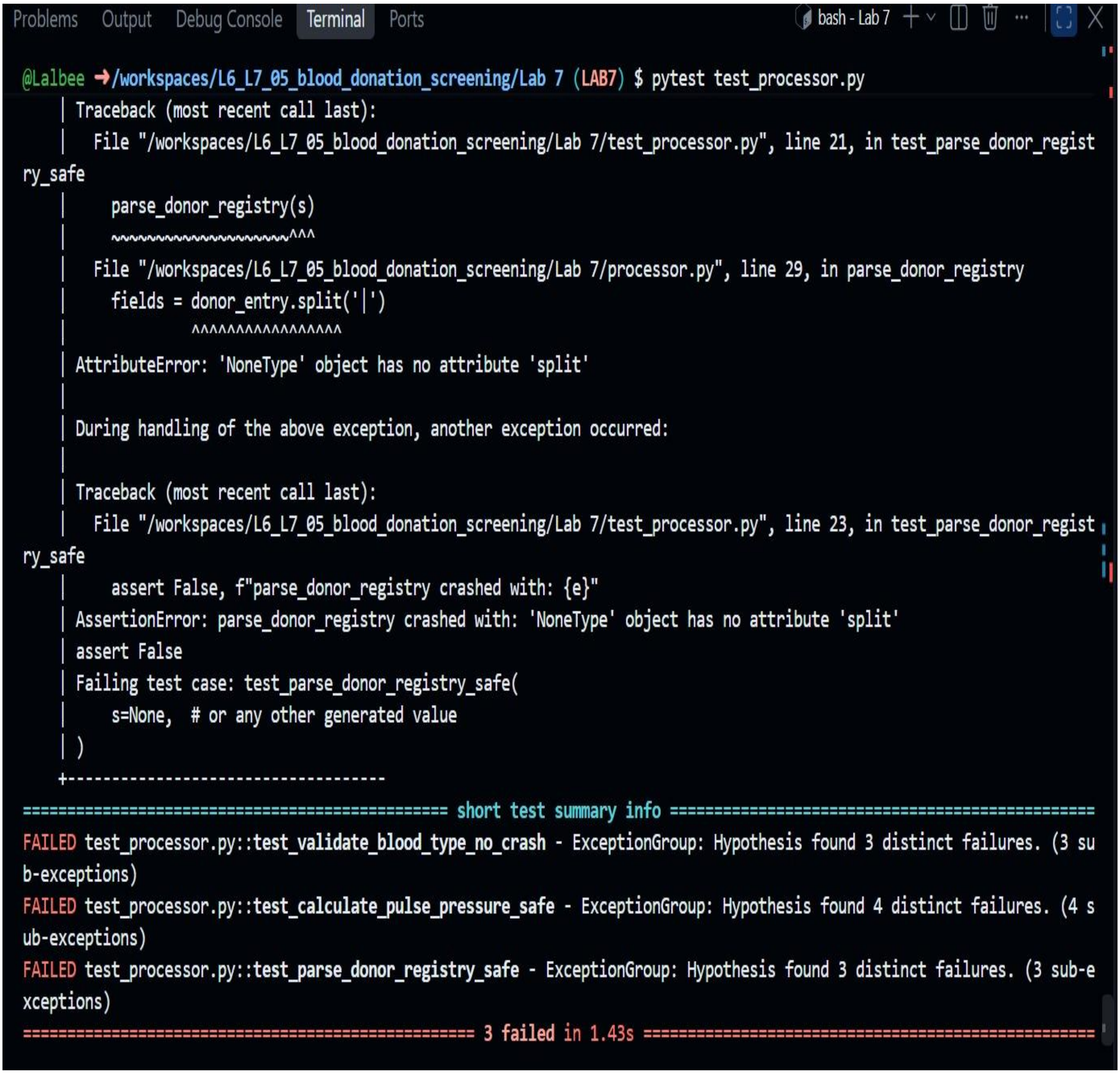
SEC:E

“Step 2: Output of buggy processor.py”



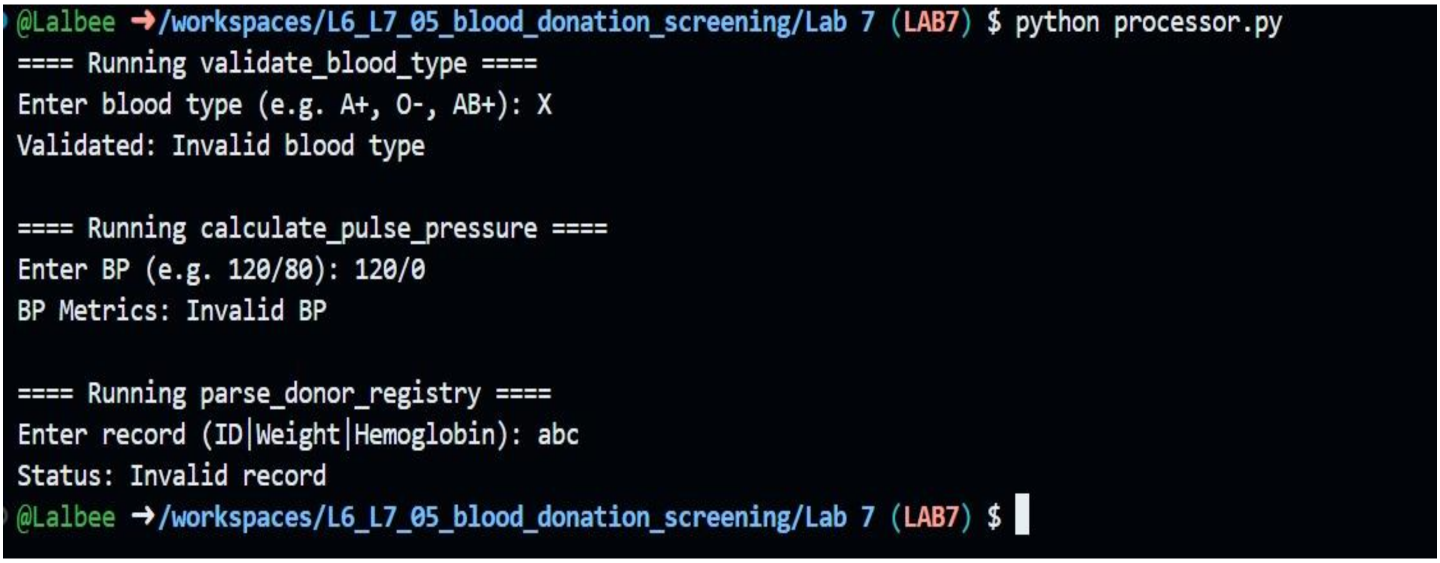
```
@Lalbee →/workspaces/L6_L7_05_blood_donation_screening (LAB7) $ cd "Lab 7"
@Lalbee →/workspaces/L6_L7_05_blood_donation_screening/Lab 7 (LAB7) $ python processor.py
==== Running validate_blood_type ====
Enter blood type (e.g. A+, O-, AB+): O+
Validated: O Positive

==== Running calculate_pulse_pressure ====
Enter BP (e.g. 120/80): 120/80
BP Metrics: Diff: 40, Ratio: 1.5

==== Running parse_donor_registry ====
Enter record (ID|Weight|Hemoglobin): D101|65|13.5
Status: Donor D101: Eligible
@Lalbee →/workspaces/L6_L7_05_blood_donation_screening/Lab 7 (LAB7) $ python processor.py
==== Running validate_blood_type ====
Enter blood type (e.g. A+, O-, AB+): x
Traceback (most recent call last):
  File "/workspaces/L6_L7_05_blood_donation_screening/Lab 7/processor.py", line 50, in <module>
    main()
    ~~~~^^
  File "/workspaces/L6_L7_05_blood_donation_screening/Lab 7/processor.py", line 39, in main
    print("Validated:", validate_blood_type(b_input))
                        ~~~~~~~~~~~~~~~~~~~^^^^^^^^^
  File "/workspaces/L6_L7_05_blood_donation_screening/Lab 7/processor.py", line 10, in validate_blood_type
    return f"{antigen} {rh_map[rh_char]}"
                        ~~~~~~^^^^^^^^^
KeyError: 'X'
```

“Step 4: pytest on buggy code (3 failed)”



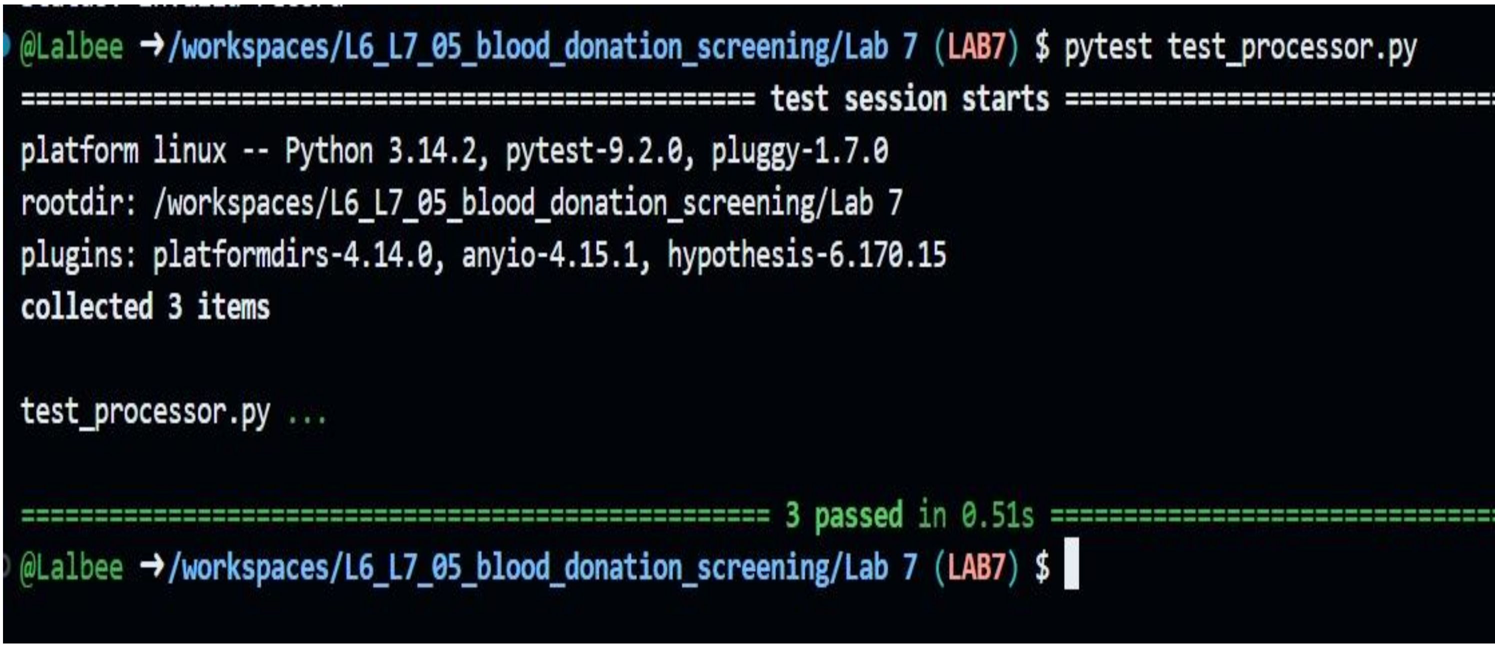
```
@Lalbee ➜ /workspaces/L6_L7_05_blood_donation_screening/Lab 7 (LAB7) $ pytest test_processor.py
    | Traceback (most recent call last):
    |   File "/workspaces/L6_L7_05_blood_donation_screening/Lab 7/test_processor.py", line 21, in test_parse_donor_regist
ry_safe
    |     parse_donor_registry(s)
    |     ~~~~~~~~~~~~~~~~~~~~^^^
    |   File "/workspaces/L6_L7_05_blood_donation_screening/Lab 7/processor.py", line 29, in parse_donor_registry
    |     fields = donor_entry.split('|')
    |              ^^^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/workspaces/L6_L7_05_blood_donation_screening/Lab 7/test_processor.py", line 23, in test_parse_donor_regist
ry_safe
    |     assert False, f"parse_donor_registry crashed with: {e}"
    | AssertionError: parse_donor_registry crashed with: 'NoneType' object has no attribute 'split'
    | assert False
    | Failing test case: test_parse_donor_registry_safe(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
================================================ short test summary info ================================================
FAILED test_processor.py::test_validate_blood_type_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 su
b-exceptions)
FAILED test_processor.py::test_calculate_pulse_pressure_safe - ExceptionGroup: Hypothesis found 4 distinct failures. (4 s
ub-exceptions)
FAILED test_processor.py::test_parse_donor_registry_safe - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-e
xceptions)
================================================== 3 failed in 1.43s ==================================================
```

"Step 5: Output of fixed processor.py"



```
@Lalbee ➜ /workspaces/L6_L7_05_blood_donation_screening/Lab 7 (LAB7) $ python processor.py
==== Running validate_blood_type ====
Enter blood type (e.g. A+, O-, AB+): X
Validated: Invalid blood type

==== Running calculate_pulse_pressure ====
Enter BP (e.g. 120/80): 120/0
BP Metrics: Invalid BP

==== Running parse_donor_registry ====
Enter record (ID|Weight|Hemoglobin): abc
Status: Invalid record
@Lalbee ➜ /workspaces/L6_L7_05_blood_donation_screening/Lab 7 (LAB7) $ 
```

“Step 6: pytest on fixed code (3 passed)”



```
@Lalbee →/workspaces/L6_L7_05_blood_donation_screening/Lab 7 (LAB7) $ pytest test_processor.py
================================================== test session starts ============================
platform linux -- Python 3.14.2, pytest-9.2.0, pluggy-1.7.0
rootdir: /workspaces/L6_L7_05_blood_donation_screening/Lab 7
plugins: platformdirs-4.14.0, anyio-4.15.1, hypothesis-6.170.15
collected 3 items

test_processor.py ...

=================================================== 3 passed in 0.51s ============================
@Lalbee →/workspaces/L6_L7_05_blood_donation_screening/Lab 7 (LAB7) $ 
```